Using density forecasts to measure the stability of inflation expectations*

Ian Dew-Becker, Stefano Giglio, and Pooya Molavi

June 9, 2026

Abstract

The anchoring of inflation expectations is a major objective of modern monetary policy. Under very weak conditions, allowing for a wide range of deviations from rationality and from the usual linear-Gaussian structure, the local sensitivity of an agent's expectations to news is equal to their subjective uncertainty multiplied by the precision of the observed signal. Instrumental variables estimates show that uncertainty is a significant determinant of expectational anchoring for survey participants, and the data is consistent the model's implication of proportionality – a hypothetical agent with zero uncertainty would ignore all signals. At the aggregate level, average uncertainty predicts the sensitivity of average expectations to realized inflation surprises, and with a magnitude consistent with the micro IV-based estimates. The results imply that as of 2025, household inflation expectations were 2–3 times more sensitive to news than prior to 2020.

One of the primary goals of modern monetary policy is ensuring that inflation expectations remain well anchored in the sense that they are not too sensitive to news and therefore do not fluctuate very much. One can think of backward-looking, contemporaneous, and forward-looking measures of anchoring. A backward-looking measure would simply ask whether inflation expectations were insensitive to news and stable over some period. Contemporaneously, the question is how sensitive expectations are to news today: what is the slope of the function mapping today's signals to expected future inflation? A forward-looking measure then would try to forecast that slope in the future.

*Dew-Becker: Federal Reserve Bank of Chicago; Giglio: Yale University; Molavi: Northwestern University. The views in this paper are those of the authors and do not represent those of the Federal Reserve Bank of Chicago. We appreciate discussions and comments from Jeff Campbell, Naveen Gondhi, and participants at the INSEAD finance symposium.

The core difficulty is that we only observe expectations in the data conditional on the news that was actually realized. Without further assumptions, it is not possible to estimate the slope of the response of expectations to news only having observed the actual news that was realized.[1] This paper's goal is to ask how well anchored expectations are by measuring, *in real time*, how sensitive expectations are to pieces of information – such as inflation announcements – holding their content (i.e. the size of the surprise) fixed.

Contribution

Measuring how inflation expectations will react to future news necessarily requires estimating a counterfactual or hypothetical: how people would change their expectations depending on what actually happens to inflation. To estimate responsiveness, this paper imposes a narrow structural assumption, which is simply that agents update their expectations using Bayes' formula and they assume their signals about future inflation have normally distributed errors. Note, critically, the assumption *is not* that they are fully rational Bayesians or that the signals are truly normal – a wide range of behavioral biases are allowed within the analysis including excessive anchoring on priors, agents having misspecified beliefs, and diagnostic expectations. It is well known that when agents are Bayesian with normally distributed priors and observe normal signals, the sensitivity of their expectations to signals is proportional to their current uncertainty (the variance of their prior). We show that that result holds locally under much more general conditions, motivating the use of uncertainty as a general measure of the strength of the expectational anchor.

The empirical analysis has two components: estimation and testing. Estimation corresponds to measuring the quantitative relationship between anchoring and reported uncertainty. By testing we mean evaluating the model's prediction that anchoring is *proportional* to uncertainty – that the intercept is zero, so that a hypothetical person with zero uncertainty has perfectly anchored expectations. We estimate and test the model in both micro and aggregate data.

Methods and results

The paper's approach is to model inflation expectations as coming from a filtering problem in which agents observe signals about future inflation, which is the latent state (a recent example of such a setup is Carvalho, Eusepi, Moench and Preston (2023)). If agents act under the assumption that signals are Gaussian (which can be motivated, for example, by assuming that they observe a large number of weakly informative signals with arbitrary distributions, or that they use normality as a simple rule of thumb), the Bayesian update

[1] See Ball and Mazumder (2011) for a discussion of anchoring in terms of either levels of expectations being stable over time or being insensitive to shocks. See also Kumar, Afrouzi, Coibion and Gorodnichenko (2015) and the discussion in Armantier et al. (2022)

step in the filter – the mapping from signals to expectations – has a relatively tractable form. Dytso, Poor and Shitz (2022) show how to express a single Bayesian update as a power series, and Dew-Becker, Giglio and Molavi (2026) show how to implement the filter dynamically.

We use two key filtering results. First, again, the local sensitivity of expectations to signals is equal to uncertainty multiplied by signal precision. Second, the evolution of uncertainty itself depends on signal precision and also on the realization of the signals, which helps formally characterize the sources of endogeneity in uncertainty.

Those results depend on the assumption that agents apply some form of Bayes' formula under the belief that their signals are normally distributed. That assumption is certainly not literally true, and it might not even be approximately true, so the paper's next task is to evaluate how well the model's predictions for sensitivity describe the data. While there are a number of data sources with information on inflation expectations, the paper focuses primarily on the New York Fed's *Survey of Consumer Expectations* (SCE). The SCE is particularly useful here because it has data available at the monthly frequency and asks respondents to provide distributions for future inflation outcomes. The main drawback of the SCE, on the other hand, is that the inflation forecasts for which there is a long time series are only at the one- and three-year horizons, whereas policymakers often focus more on longer-term expectations.

In the cross-section of the SCE, there turns out to be if anything a *negative* relationship between uncertainty and responsiveness to signals. That result is not surprising since uncertainty is endogenous to signal precision. Agents who consume a lot of information and therefore have precise signals will also tend to have the lowest uncertainty, making the reduced-form relationship between sensitivity and uncertainty ambiguous.

We therefore use the panel dimension of the data to run an instrumental variables analysis. Kim and Binder (2023) show uncertainty systematically declines with tenure in the survey and we give conditions under which it is a valid instrument. The decline in uncertainty is consistent with people on average paying more attention to inflation while they are in the survey than they did previously. It is natural to think that after being asked many questions about inflation, news about it would be more salient, and the questions in the survey themselves contain implicit information about inflation (e.g. its typical scale). The exogeneity assumption is that attention (effective signal precision) is orthogonal to tenure – that is, while people pay more attention while in the survey than they did previously, we need the restriction that their attention does not decline over the course of the survey. The fact that uncertainty declines monotonically suggests that attention remains at least somewhat stable across months of tenure. The availability of this instrument is another feature of the SCE that distinguishes it from other surveys, such as the Survey of Professional Forecasters.

In the cross-section of the SCE, the reduced-form finding is that the survey respondents with low tenure have relatively high uncertainty and their expectations both covary more strongly with inflation surprises and are more volatile overall than those of respondents with high tenure, consistent with the model. Under the instrumental variables interpretation, then, the second stage says that high uncertainty is associated with larger responses to inflation surprises and greater volatility in beliefs: respondents with low tenure have expectations that are more weakly anchored than those with high tenure under both the level and shock concepts of Ball and Mazumder (2011).

The quantitative relationship between responsiveness to news and uncertainty – a slope – measures in the model agents' average signal precision. The model has an additional prediction, though, which is that an agent with zero uncertainty should have zeor responsiveness – the intercept is zero. We show that the data does not reject that null.

We also examine time-series regressions using average uncertainty and inflatoin expectations. The magnitude of the aggregate time-series relationship between average uncertainty and the sensitivity of average expectations to inflation is consistent with the micro IV-based estimates, and we again get the result that the estimated intercept is zero.

Related work

Armantier et al. (2016) provide closely related evidence from an information provision experiment that agents are approximately Bayesian in the sense that their responsiveness to information increases in prior uncertainty.[2] Related work includes Weber et al. (2025), Coibion and Gorodnichenko (2015), and Cavallo et al. (2017). Note that those papers work in rational linear-Gaussian setups, which the theoretical structure here generalizes significantly.

Another complementary approach is Armantier et al. (2022), which directly asks people about how their expectations would change in response to certain specific events, such as particular realizations of inflation or unemployment.[3]

Lamla and Vinogradov (2019) and Binder, Campbell and Ryngaert (2024) also study responses of inflation expectations in the SCE, focusing on within-month dynamics (using the fact that the SCE reports the exact date on which each response was recorded). Our analysis implies that the high-frequency responsiveness that they measure could be further conditioned on uncertainty (or tenure in the survey).

[2] See also Weber et al. (2025).

[3] Ameriks et al. (2011) introduce the idea of strategic surveys as a solution to the sort of identification problem that this paper faces. When counterfactuals are not observable, the approach is to simply ask about them. Armantier et al. (2015) and Roth and Wohlfart (2020) among others provide evidence that people act on their beliefs.

Carvalho et al. (2023) is also very closely related. It posits a New Keynesian structure for inflation dynamics and then assumes that agents form expectations through a rule of thumb filtering rule. The inflation process in that paper is one particular case of the general class of processes allowed in this paper's analysis. The primary contrast is that this paper assumes that agents are Bayesian or quasi-Bayesian, whereas Carvalho et al. (2023) assumes they are boundedly rational. That paper also estimates a fully specified model for inflation and expectations, whereas the nature of the more general setup here means that it does not yield a full description of dynamics.

Given this paper's focus on the conditional distribution of future inflation, it is naturally also related to work that studies option-implied distributions, including Kitsul and Wright (2013), Fleckenstein, Longstaff and Lustig (2017), Mertens and Williams (2021), and Hilscher, Raviv and Reis (2025). The basic implication of this paper's analysis is that option-implied uncertainty would also be a measure of expectational anchoring, and studying that would be a straightforward extension of the analysis.

More generally, Reis (2025) reviews recent evidence on inflation expectations and their link to post-Covid inflation. This paper builds closely on that work by asking what factors drive expectations. Reis (2025) emphasizes the causal effects that expectations can have. The evidence this paper presents that expectations are now less firmly anchored than prior to Covid then suggests that inflation itself may in the future be more volatile. See also D'Acunto and Weber (2024) and Coibion and Gorodnichenko (2025).

Outline

Section 1 develops the paper's core economic structure and the link between posterior beliefs and the response function for expectations. Section 2 examines how the theory can be tested empirically and section 3 provides estimates. Finally, section 4 studies the model's implications for forecasting expectational anchoring going forward and section 5 concludes.

1 Theory

The aim of this section is to characterize the drivers of expectations under relatively weak assumptions about priors and rationality. Specifically, given a signal y_t, with π_t^* being the object agents are learning about, we assume that agents have some subjective probability distribution with a date-t mean and variance,

$$m_t \equiv E\left[\pi_t^* \mid y^t\right] \tag{1}$$

$$u_t \equiv \mathrm{var}\left(\pi_t^* \mid y^t\right) \tag{2}$$

where y^t denotes the history of observed signals up to date t. There is no assumption yet about the rationality of the probability distribution or how it is updated over time. π_t^* can correspond to either trend inflation or actual realized future inflation.

1.1 Basic updating result

1.1.1 Economic environment

Assumption 1 *π_t^* follows an arbitrary dynamic process. In particular, it is not necessarily linear, Gaussian, or homoskedastic.*

Given the observed dynamics of inflation, assuming π_t^* is normal here would fail to fit many features of the data. In particular, inflation is positively skewed (the skewness coefficient for core PCE inflation is 1.24) and its volatility varies significantly over time (e.g. Engle (1982) and Stock and Watson (2007)).

Assumption 2 *Agents receive a signal y_t about π_t^* which they assume (possibly incorrectly) is distributed as*

$$y_t \sim N\left(\pi_t^*, \sigma_t^2\right) \tag{3}$$

σ_t contains no information about future inflation beyond what is contained in y_t itself.

In practice the signal is agent-specific, and below we add an i subscript to indicate different agents. It is also straightforward to generalize the analysis to allow multiple signals.

One interpretation of the parameter σ_t^{-2} is that it measures how much attention agents pay to inflation. High attention corresponds to having signals with high precision. It may also depend on the amount of information available to agents.

The assumption that agents treat the signals as conditionally normally distributed is somewhat restrictive (if very standard), but necessary for the main theorem we use below.[4] Note, though, that we neither assume that the signal is *truly* normally distributed nor that its mean and variance are actually π_t^* and σ_t^2.[5]

[4]One way to motivate it is to assume that agents receive many independent signals centered on π_t^*, each with very low precision, that have arbitrary distributions. Then the martingale central limit theorem implies that the sum of the signals (i.e. y_t) is asymptotically – as the precision goes to zero and the number of signals goes to infinity at the same rate – normally distributed and the sufficient statistic for the Bayesian update (Hall and Heyde (2014)). Alternatively, agents might truly receive normally distributed signals, or, again, they might assume normality as a simple rule of thumb.

There are other versions of the main result we use under alternative distributional assumptions for the signal. Normality is the usual baseline, but in other settings a different distribution (e.g. Poisson) could be natural.

[5]It is also worth noting that we are assuming that the errors in the signal are independent of π_t^*. Dew-Becker, Giglio and Molavi (2026) show how the results change when there is feedback from beliefs to the latent process π_t^*.

1.1.2 Bayes' formula and rationality

Assumption 3 *On all dates agents have subjective probability distributions over potential values of π_t^*. The distributions are updated using Bayes' formula based on their beliefs about the dynamic process driving π_t^* and the signal distribution (assumption 2).*

Assumption 3 says that agents are rational to the extent that their beliefs can be described by probability distributions that they update via Bayes' formula. It does not impose, though, that their specification for either the signals (assumption 2) or for the dynamics of π_t^* (assumption 1) is correct. Specifically, we can have the following deviations from rationality:

1. **Signal distribution:** Agents may use an incorrect value for signal precision (σ_t^{-2}) in calculating their update. They might place too much or too little weight on their signals (e.g. as in diagnostic expectations) or ignore some sources of information. The signals might also not actually be Gaussian.

2. **Prior distribution:** Agents' prior need not be correct in any way. It might be wrong if agents have the wrong dynamic model for inflation, they might have some strange prior based on superstition, or they might just for some reason have started with beliefs far from the truth (e.g. Farmer, Nakamura and Steinsson (2024)). The prior need not even be absolutely continuous with respect to any sort of true distribution, and agents need not have common priors (e.g. as in Malmendier and Nagel (2016)).

What we impose is simply that agents act as though their signals have normally distributed errors and that they update using Bayes' formula. Many behavioral models are therefore consistent with the paper's analysis, including Daniel et al. (1998), DeMarzo et al. (2003) (see Molavi (2025)). It is also straightforward to show that the paper's results hold under diagnostic expectations (as in Bordalo et al. (2019)), with the behavioral bias appearing as an effective rescaling of the signal precision.

1.1.3 The update equation

It turns out that the standard updating formula for mean expectations that holds in the fully Gaussian case continues to hold here. The simplest way to do that derivation is via a discrete-time approximation from the results of Dew-Becker, Giglio and Molavi (2026).

Proposition 1 *Under assumptions 1–3, in the limit when time periods are short or signals are uninformative, expectations are updated according to*

$$\Delta m_t \approx u_t \sigma_t^{-2} \left(y_t - m_{t|t-1} \right) + E_{t-1} \left[\Delta m_t \right] \tag{4}$$

where $m_{t|t-1} \equiv m_{t-1} + E_{t-1} \left[\Delta m_t \right]$.

(4) is the main equation that the paper aims to test. The true update is nonlinear, but to first order the slope is equal to agents' uncertainty multiplied by their belief about signal precision. Equation (4) is useful not because it implies novel dynamics for beliefs, but rather because it shows that the standard formula in fact holds under very general conditions. Its key implication is that the sensitivity of beliefs to signals is equal to uncertainty multiplied by signal precision. In principle the approximation can be written in terms of either u_t or u_{t-1} – the difference is second-order (and in continuous time they are equal). We write it in terms of u_t because that maps directly onto the analogous discrete-time result (see Dytso, Poor and Shitz (2022)). Formally, $\partial m_t / \partial y_t = u_t \sigma_t^{-2}$.

Figure 1 displays an example of the posterior expectation as a function of the signal. The mapping is nonlinear, but its slope at each point is equal to the posterior second moment for that value of the signal multiplied by signal precision. In this example, for relatively low signals, the posterior variance is small and the posterior expectation is relatively insensitive to signals. For high signals, the posterior variance is high and the expectation is much more sensitive.

1.1.4 The effect of signal precision on uncertainty

Uncertainty is not itself exogenous to σ_t^{-2} – both agents' signal precision and the signal realizations themselves affect uncertainty. We formalize that again with a discretization of a result from Dew-Becker, Giglio and Molavi (2026).

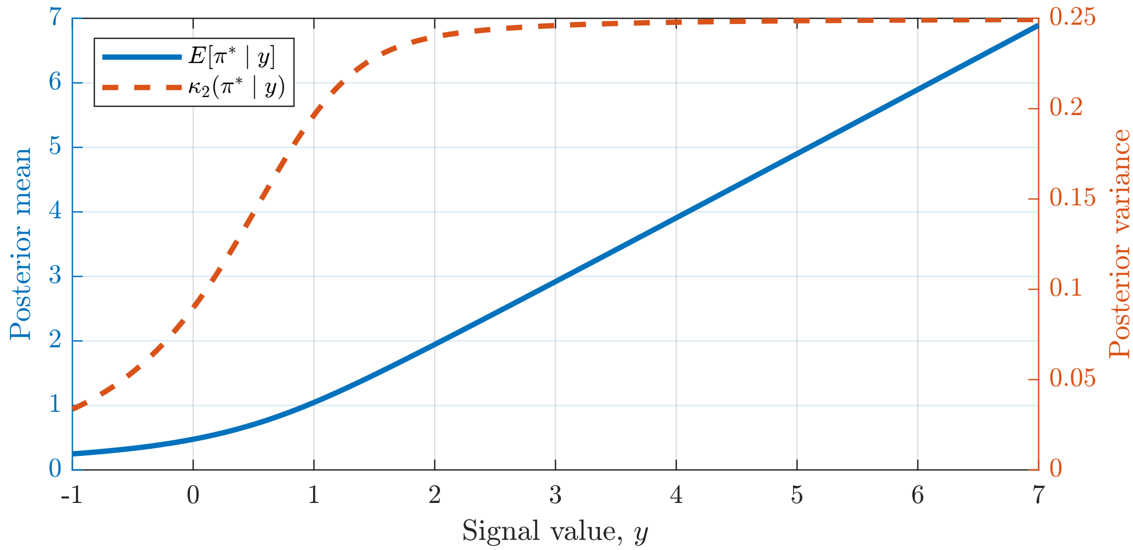
Proposition 2 *Uncertainty updates according to*

$$\Delta u_t \approx \mathrm{var}_{t-1} \left(\Delta \pi_t^* \right) + 2 \,\mathrm{cov}_{t-1} \left(\Delta \pi_t^*, \pi_t^* \right) - \sigma_t^{-2} u_{t-1}^2 + \sigma_t^{-2} \mu_{3,t} \left(y_t - m_{t|t-1} \right) \tag{5}$$

where $\mu_{3,t}$ *denotes the third central moment of the posterior distribution of* π_t^*.

As with proposition 1, proposition 2 is a generalization of the Riccati equation for uncertainty that holds in the fully Gaussian case. The first two terms represent the increase in uncertainty from shocks to fundamentals. The third term is the decline coming from information accumulation and the fourth is the effect of signals on uncertainty.

Figure 1: Hypothetical Bayesian update



Note: The figure assumes a prior distribution $\pi^* \sim \chi^2(3)$ and that $y \sim N(\pi^*, (1/2)^2)$. The lines plot the posterior mean and variance for π^* conditional on the observed signal.

There are two important implications of proposition 2 for the analysis here. First, we have the standard result that high precision in signals reduces uncertainty (the third term), so that u and σ^{-2} should be negatively correlated across people.

Second, the generalization to arbitrary priors reveals, through the last term, that u is also endogenous to the actual realization of the signals with a coefficient proportional to the third moment (again, the posterior is used because, formally, $\partial u_t / \partial y_t = \sigma_t^{-2} \mu_{3,t}$). When beliefs are asymmetrical, signals affect uncertainty by determining whether agents think the longer or shorter tail of the distribution is more likely to be realized. For example, when the subjective distribution is positively skewed, a high inflation signal makes the right tail, which is the longer one, more likely and raises subjective uncertainty.

The dependence of uncertainty on both signal precision and realized signals will be major complications that need to be accounted for in the empirical analysis.

1.2 Observable predictions

This section derives the key equations that we estimate empirically. It extends the analysis above to explicitly account for cross-sectional heterogeneity, with agents indexed by i.

1.2.1 Squared changes in expectations

Squaring equation (4), taking expectations, and ngoring higher order terms, we have

Corollary 1 *For an individual agent*

$$E_{t-1}\left[(\Delta m_t)^2\right] \approx \left(E_{t-1}\left[u_t\right]\right)^2 \sigma_t^{-2} \tag{6}$$

ignoring higher order terms in σ^{-2}.

Corollary 1 says that squared changes in beliefs are on average equal to squared (expected) uncertainty multiplied by signal precision. In principle a regression of squared changes in expectations on squared uncertainty should potentially yield an estimate of average signal precision. We analyze identification conditions in the next section.

Corollary 1 also implies thtat the coefficient in a regression of volatility in expectations on squared uncertainty should have an intercept of zero, which is a prediction that yields a test of the model.

One potential concern with a regression for squared changes in expectations is that there might be noise in reported expectations and that the magnitude of the noise could be correlated with tenure or uncertainty. We will argue below that the data appears consistent with that. The regression with squared changes has the advantage of requiring fewer structural assumptions than the analysis in the next section, but the drawback that it can be biased by certain patterns in reporting error.

1.2.2 Responses of expectations to realized inflation

While we do not know all the information agents use in forecasting inflation, it is natural to think that realized inflation is one component. To rationalize that and also impose some structure to help in estimation, a natural special case of the model is the trend-plus-noise specification for inflation.

Definition 1 ***Trend plus noise model.*** π_t^* *represents trend inflation. Realized inflation,* π_t*, follows*

$$\pi_t = \pi_t^* + \varepsilon_{\pi,t}, \text{ with } \varepsilon_{\pi,t} \sim N\left(0, \sigma_{\pi,t}^2\right) \tag{7}$$

Agents observe a noisy signal about realized inflation, $s_{i,t}$ *(consistent with limited attention or with people just observing a subset of prices):*

$$s_{i,t} = \pi_t + \varepsilon_{s,i,t}, \text{ with } \varepsilon_{s,i,t} \sim N\left(0, \sigma_{s,i,t}^2\right) \tag{8}$$

Agents also observe a second signal, $x_{i,t}$, that is independent of $s_{i,t}$ conditional on π_t^* and has an arbitrary distribution. $x_{i,t}$ represents all the other information agent i gets about trend inflation (e.g. from policy announcements) and acts as a residual.

In the trend plus noise model, proposition 1 can be adjusted to get the following result.

Corollary 2 *Define $\hat{\pi}_{i,t} \equiv \pi_t - m_{i,t-1}$ and $\sigma_{i,t}^{-2} \equiv \left(\sigma_{s,i,t}^2 + \sigma_{\pi,t}^2\right)^{-1}$. Then in the trend plus noise model*

$$\Delta m_{i,t} = u_{i,t}\sigma_{i,t}^{-2}\left(\pi_t - m_{i,t-1}\right) + \varepsilon_{i,t} \tag{9}$$

$$\text{where } \varepsilon_{i,t} \equiv u_{i,t}\sigma_{i,t}^{-2}\left(s_{i,t} - \pi_t\right) + f_{i,t-1}\left(x_{i,t}\right) + \text{higher order terms} \tag{10}$$

$f_{t-1}(\cdot)$ is a function representing how expectations are updated given the unobservable signal $x_{i,t}$.[6] *Up to terms of order $\sigma_{i,t}^{-2}$ or smaller, $u_{i,t}\sigma_{i,t}^{-2}\left(\pi_t - m_{i,t-1}\right)$ and $\varepsilon_{i,t}$ are both mean zero under the agent's probability measure.*

So in the trend plus noise model, a natural regression would now be of changes in expectations on uncertainty multiplied by the inflation surprise.

As with squared changes in expectations, the model also implies a particular zero restriction, which is that the sensitivity to inflation surprises should be zero for a hypothetical agent with zero uncertainty.

1.2.3 The sensitivity of mean expectations to realized inflation

The paper's core objective is to understand the sensitivity of expectations to signals. Clearly at any given time, the average sensitivity is the average of $u_{i,t}\sigma_{i,t}^{-2}$. We argue below that uncertainty is observable from survey data (and prices of financial market instruments). To see why that matters, consider a shock that raises uncertainty of all agents by some amount Δ_u, leaving signal precision fixed:

$$\frac{\partial}{\partial \Delta_u}\frac{\partial Avg_t\left(m_{i,t}\right)}{\partial \pi_t} = Avg_t\left(\sigma_{i,t}^{-2}\right) \tag{11}$$

where $Avg_t(\cdot)$ denotes the cross-sectional average on date t.[7] In words, an exogenous shift in uncertainty raises the sensitivity of expectations to realized inflation – and, more generally, to any signal – in proportion to the average signal precision. That average is fundamentally our

[6]Specifically, $f_{t-1}(x_t) = E\left[\pi_t^* \mid y^{t-1}, x_t, \hat{s}_t = 0\right] - E\left[\pi_{t-1}^* \mid y^{t-1}\right]$.

[7]Equation (11) follows directly from results in Dytso, Poor and Shitz (2022) and is exact, rather than being a derivative of the approximation (9).

parameter of interest when estimating (rather than testing) the model. Equation (11) shows formally that really we are after a second-order sensitivity – the response of the sensitivity of expectations to a change in uncertainty.

Equation (11) essentially gives an impulse response. In practice one might naturally be concerned that the cross-sectional means of expectations, uncertainty, and signal precision are all jointly determined. The empirical analysis will focus on partial effects, such as that in (11), but also study a simpler forecasting problem that avoids the entire question of exogeneity and identification and just asks whether reported uncertainty has predictive power for future movements in expectations.

2 Empirical identification

This section discusses what assumptions are needed in order to estimate the regressions suggested by the theoretical analysis. The key complication is the dependence of uncertainty and signal precision, which we address with instrumental variables methods. We begin with the trend-plus-noise model because the analysis is simpler without dealing with squared variables.

Throughout the discussion, the assumption is that uncertainty, $\mathrm{var}_{i,t}(\pi_t^*)$, is observable, while signal precision, $\sigma_{i,t}^{-2}$, is not.

2.1 Identifying assumptions

To avoid the endogeneity of uncertainty to both $\sigma_{i,t}^{-2}$ and $\varepsilon_{i,t}$ (from 9), we instrument for uncertainty. Motivated by the finding in Kim and Binder (2023) that uncertainty declines on average with tenure in the survey, we use tenure as the instrument.

Because the endogeneity here comes in an interaction, two-stage least squares in general need not yield a consistent estimator of the mean of $\sigma_{i,t}^{-2}$. While we report results for 2SLS, the primary results are for a control function estimator, which Wooldridge (2015) and Masten and Torgovitsky (2016) argue is preferable in settings such as ours with heterogeneous slopes and in which the endogenous variable enters as an interaction.

The analysis still centers on the usual exogeneity and relevance assumptions, but stated in a slightly different way from in 2SLS:

Assumption 4 *[Relevance] There is a first-stage representation for uncertainty, $u_{i,t}$,*

$$u_{i,t} = a_u + b_u z_{i,t} + \eta_{i,t}, \tag{12}$$

where tenure, $z_{i,t}$, is independent of the residual $\eta_{i,t}$, $b_u \neq 0$, and

$$\operatorname{var}\left(z_{i,t} \mid \eta_{i,t}, \hat{\pi}_{i,t}\right) > 0 \tag{13}$$

Note that in (12) $\eta_{i,t}$ may be correlated within i, for example if there are person-level fixed effects.

Assumption 5 *[Exogeneity] Tenure, $z_{i,t}$, is exogenous and uncorrelated with $\sigma_{i,t}^{-2}$ in the sense that*

$$\mathbb{E}\left[\sigma_{i,t}^{-2} \mid z_{i,t}, \eta_{i,t}, \hat{\pi}_{i,t}\right] = \mathbb{E}\left[\sigma_{i,t}^{-2} \mid \eta_{i,t}, \hat{\pi}_{i,t}\right] \tag{14}$$

$$\mathbb{E}\left[\varepsilon_{i,t} \mid z_{i,t}, \eta_{i,t}, \hat{\pi}_{i,t}\right] = \mathbb{E}\left[\varepsilon_{i,t} \mid \eta_{i,t}, \hat{\pi}_{i,t}\right] \tag{15}$$

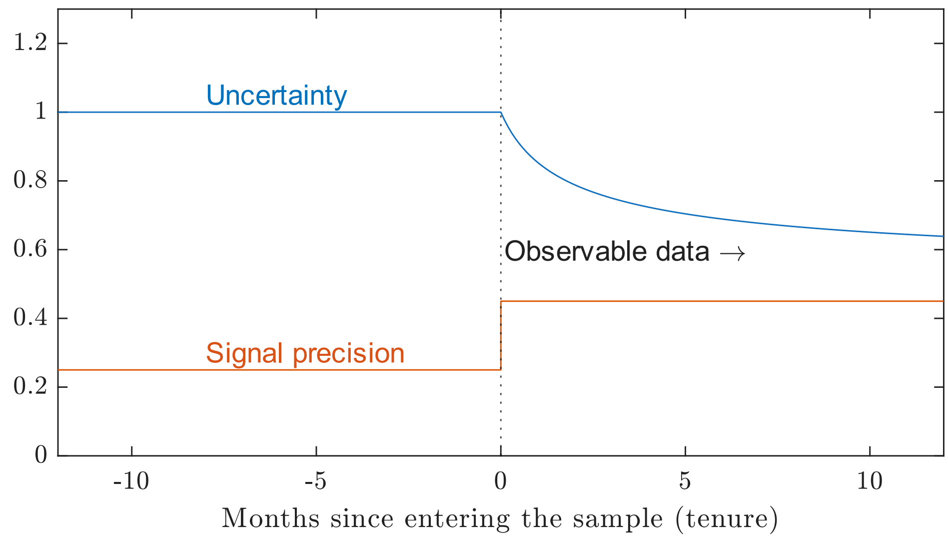
where $\mathbb{E}$ denotes the population expectation operator.

One point to note is that the relevance assumption is stricter than usual, in the sense that we need a functional form along with exogeneity to the first-stage residual. Linearity is not by itself important, but in order to ensure identification we do need a structural interpretation to the first stage in which the residual is exogenous. It will be important both in interpreting two-stage least squares and an alternative control function estimator (see Wooldridge (2015) and Masten and Torgovitsky (2016) for discussions of the first stage assumption).

The first economic question is what would create relevance here. Why would tenure be related to uncertainty? Empirically, Kim and Binder (2023) show, and we replicate below, that uncertainty declines with tenure on average. Kim and Binder (2023) discuss how that can be expected to happen when people pay more attention to inflation news when they are in the survey than they did previously, including potentially learning from the survey itself. In terms of the model, that corresponds to signal precision – $\sigma_{s,i,t}^{-2}$ – increasing on entry into the survey, which then leads $u_{i,t}$ to decline over time via equation (5). In other words, the treatment here is the survey itself.

Figure 2 visualizes the basic idea behind the identification. The x-axis represents the number of months since a respondent has entered the survey, with points to the left of zero corresponding to dates prior to the survey. The key assumption is that when people enter the survey, they pay more attention to inflation, which appears here as a level shift in signal precision. That then causes their uncertainty to drift down through equation (5). The identification in the regression comes from the within-person variation, where their uncertainty falls with tenure.

Figure 2: Intuition behind the identifying assumption



Note: The figure shows signal precision and uncertainty for a hypothetical agent who satisfies the identifying assumption. They enter the survey at point 0 on the x-axis. The blue line represents their uncertainty, $u_{i,t}$, and the orange line the precision of their signals ($\sigma_{s,i,t}^{-2}$) – how much attention they pay to inflation. In the period that they are in the survey, the assumption is that attention is unrelated to tenure (though need not be constant), leading uncertainty to progressively decline.

The figure also helps to see what factors could cause a bias. One would be if attention varies systematically with tenure. The identifying assumption is that attention is higher when a person is in the survey, but that it is uncorrelated with tenure during the survey. If attention rises with tenure on average, that would counteract the effect of uncertainty and bias our estimates towards zero, whereas if it falls, that would compound the effect of uncertainty and cause us to overestimate the dependence of updating on uncertainty. Note that the assumption is not that signal precision is constant; it just needs to be on average unrelated to tenure (formalized in assumption 5). While attention obviously could systematically vary with tenure, it is not obvious whether it would rise or fall, and on average across people it might be constant. Furthermore, even if there are differences across tenure, the first-order point is simply that being in the survey is very different from not being in it.

2.1.1 Control function estimator

The control function estimator uses the residual from the first stage, $\eta_{i,t}$ in equation (12), as a control in the second stage. In our case, since the second stage is nonlinear, $\eta_{i,t}$ also needs to be included as an interaction. The basic second-stage regression for the trend-plus-noise model is then

$$\Delta m_{i,t} = u_{i,t}\hat{\pi}_{i,t}\mathbb{E}\left[\sigma_{i,t}^{-2} \mid \eta_{i,t}, \hat{\pi}_{i,t}\right] + \mathbb{E}\left[\varepsilon_{i,t} \mid \eta_{i,t}, \hat{\pi}_{i,t}\right] + \nu_{i,t} \tag{16}$$

The two expectations play the role of the slope and intercept. Here they are left as free functions. Similarly, for the squared change in beliefs, the second stage is

$$(\Delta m_{i,t})^2 = u_{i,t}^2\mathbb{E}\left[\sigma_{i,t}^{-2} \mid \eta_{i,t}\right] + \mathbb{E}\left[\lambda_{i,t} \mid \eta_{i,t}\right] + \mu_{i,t} \tag{17}$$

The equations fit into the partially nonparametric setup analyzed by Newey and Stouli (2021), who show that the conditional expectation functions are identified (their theorem 4).

As discussed above, the fundamental parameter to be estimated is $\mathbb{E}\left[\sigma_{i,t}^{-2}\right]$. That can be estimated by averaging the conditional expectation across the values of $\eta_{i,t}, \hat{\pi}_{i,t}$. We estimate the conditional expectation functions by series least squares – i.e. approximating them with a set of basis functions (starting from a simple linear specification) – as suggested by Newey and Stouli (2021) (and see Newey (1997)).

2.1.2 Two-stage least squares

In general 2SLS does not yield consistent estimates of $\mathbb{E}\left[\sigma_{i,t}^{-2}\right]$.

Proposition 3 *In addition to the assumptions above, suppose $Z_{i,t} \equiv z_{i,t}\hat{\pi}_{i,t}$ is a valid instrument for $u_{i,t}\hat{\pi}_{i,t}$, in that $E\left[Z_{i,t}\varepsilon_{i,t}\right] = 0$. Then the population second-stage estimate from a 2SLS regression of $\Delta m_{i,t}$ on $u_{i,t}\hat{\pi}_{i,t}$ and a constant with instrument $Z_{i,t}$ is*

$$\beta^{2SLS} = \frac{E\left[\sigma_{i,t}^{-2}u_{i,t}\hat{\pi}_{i,t}\hat{Z}_{i,t}\right]}{E\left[u_{i,t}\hat{\pi}_{i,t}\hat{Z}_{i,t}\right]} \tag{18}$$

where $\hat{Z}_{i,t} = Z_{i,t} - E\left[Z_{i,t}\right]$.

As usual, the two-stage least squares estimator is a weighted average of the true treatment effect, in this case $\sigma_{i,t}^{-2}$. Note, though, that the weights here need not necessarily be positive, which complicates the interpretation. It is well known that in the case of heterogeneity in treatment effects – here, heterogeneity in $\sigma_{i,t}^{-2}$ – two-stage least squares estimates a weighted average. The average in this case is not in general the population average that we are most interested in. That said, it is entirely possible for this average to happen to be near the population average, and we report conventional 2SLS results as robustness.

3 Panel analysis

3.1 Data

We study the Federal Reserve Bank of New York's *Survey of Consumer Expectations* (SCE). The SCE is well suited to testing this paper's hypotheses both because it is a relatively large panel – with respondents remaining in the survey for up to 12 months – and because it is a representative sample of consumers, meant to capture inflation expectations of typical agents in the economy. The SCE is also run monthly, rather than quarterly, giving relatively fine time-series detail, which is useful since the most recent episode of high inflation lasted only about two years.

The SCE asks respondents about inflation expectations at horizons of 1–12, 25–36, and 49–60 months. First, it asks for point estimates: "What do you expect the rate of inflation/deflation to be over the next 12 months? Please give your best guess." Next, it asks for probabilities that inflation falls into different bins: "Now we would like you to think about the different things that may happen to inflation over the next 12 months... what would you say is the percent chance that, over the next 12 months..." followed by a list of bins for inflation: >12%, 8-12%, etc. Equivalent questions are asked for annual inflation starting 24 and 48 months in the future. We use both the point estimates and the bins.

The bins are critical because they make it possible to calculate a mean and conditional variance for each person. We use the estimates of each respondent's variance that is constructed by the administrators of the SCE, who do so by fitting a beta distribution to each respondent's reported bin probabilities.[8]

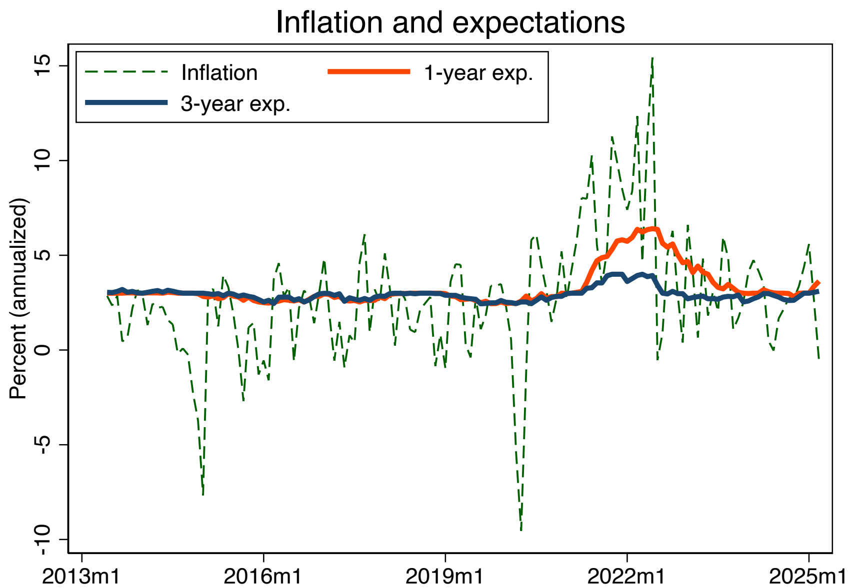
The paper's sample period is from the beginning of the SCE in June, 2013, through July, 2025. for context, figure 3 plots the time series of headline CPI inflation over that period along with 1- and 3-year inflation expectations (panel (a)) and with 1- and 3-year uncertainty (panel (b)), obtained from the SCE.

3.2 Mapping from the data to the model

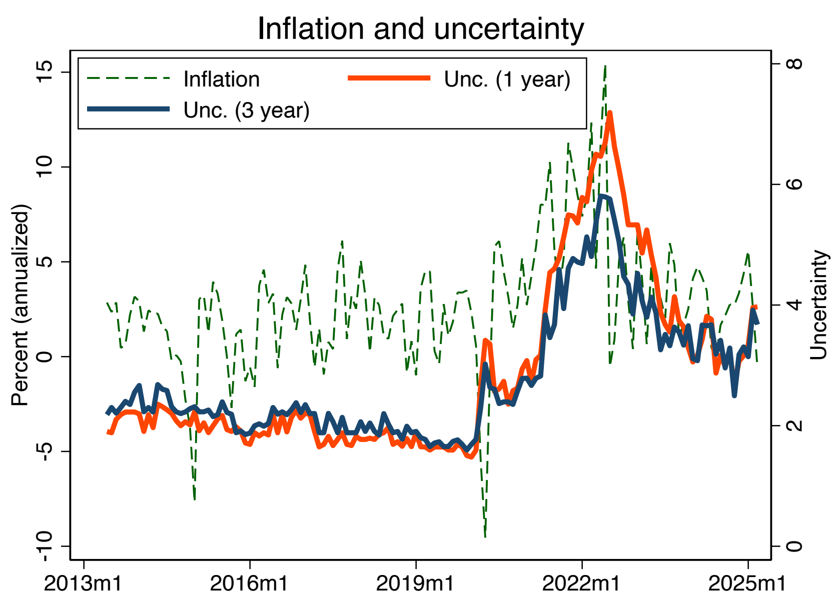
In the case where we study squared changes in expectations and do not impose the trend-plus-noise model, π_t^* in the model corresponds to future realized inflation. That is, the survey asks about realized inflation in a given 12-month period, and that random variable is π_t^* in

[8] They do not calculate a variance for responses that put positive probability on a set of non-contiguous bins, and we therefore drop those observations from the analysis. We additionally drop observations where the variance is greater than 200 squared percentage points and those with bin means outside the 5th and 95th percentiles or where the absolute change in expectations is greater than 8 percentage points.

Figure 3: Inflation, expectations, and uncertainty



(a) Inflation and expectations



(b) Inflation and uncertainty

Note: Panel (a) panel plots CPI inflation (annualized) together with the cross-sectional median 1-year and 3-year inflation expectations from the SCE. Panel (b) plots CPI inflation (left axis, annualized) together with the cross-sectional median uncertainty measures, at the 1-year and 3-year horizon on the right axis.

the model. When people report a probability distribution for future inflation, we interpret that as a probability distribution for π_t^*. Note that is entirely consistent with the setup – future inflation is a hidden random variable that follows some stochastic process (assumption 1) about which agents receive some sort of information (assumption 2) and over which they report beliefs (assumption 3).

For the trend plus noise model, the mapping is slightly more complicated. In the model, π_t^* represents trend inflation, but the survey does not ask about the trend, but rather the realization. However, given that the trend-plus-noise model assumes that the noise in inflation has zero mean, agents' reported expectation for future inflation also corresponds to an expectation of π_t^*.

Their subjective distribution over inflation, however, is not equal to their distribution for π_t^*. Instead, given equation (7), we have, at any horizon,

$$\operatorname{var}_t(\pi_{t+j}) = \underbrace{\operatorname{var}_t(\pi_t^*)}_{=u_t} + j \operatorname{var}(\Delta\pi_t^*) + \sigma_\pi^2 \tag{19}$$

where here for simplicity we are assuming that the innovations to the trend and the noise component of inflation both have constant variance and that the trend is a random walk (all typical assumptions). The left-hand side corresponds to the variance actually reported in the survey, while the first term on the right is the relevant concept of uncertainty from the model. Under the assumption that the last two terms in (19) are constant over time, agents' reported subjective uncertainty over future inflation is equal to their uncertainty about trend inflation plus a constant.

Note also that since all of the terms in (19) are nonnegative, reported uncertainty is zero only if u_t is also equal to zero, so a person who reports zero uncertainty in the trend plus noise model should have zero sensitivity to realized inflation.

Overall, consistent with the analysis above, the trend plus noise model requires more assumptions than the specification involving squared changes in expectations, but it has the advantage that it allows for analysis of the relationship between expectations and realized inflation.

3.3 Calculating inflation surprises

We calculate the inflation surprise in each month for each person as realized inflation – PCE or CPI inflation, headline or core – minus that person's expected one-year inflation – either the point forecast or the mean from the bins – in the previous month. Note that the actual CPI and PCE numbers may not have been released by the time of the survey. The idea here is rather that the BLS-measured value of inflation is just a proxy for what people observe, which is a mix of the prices they pay and other news sources.

Since the change in expectations, $\Delta m_{i,t}$ is on the left-hand side of the regression, there might be a concern about using $m_{i,t-1}$ on the right-hand side of the regression in constructing the inflation surprise (e.g. if there is measurement error in the expectations). For that reason, any time the same measure of expectations is on the right- and left-hand sides of the regression, the analysis uses the second lag of expectations in calculating the surprise. That is, for those cases $\hat{\pi}_{i,t} \equiv \pi_t - m_{i,t-2}$.

In what follows, $\hat{\pi}_{i,t}$ continues to represent the generic inflation surprise, which may be measured in any of the ways described above – there are 16 total possible permutations.[9]

In the main text, inflation is measured as the CPI headline value. The dependent variable, $\Delta m_{i,t}$ is measured based on the mean from respondents' probability distributions. Inflation surprises are then measured as realized CPI headline inflation minus lagged expectations. The main text reports results calculating the surprise relative to either the bin mean or the point forecast. Robustness to all these choices is reported in the appendix.

3.4 First stage

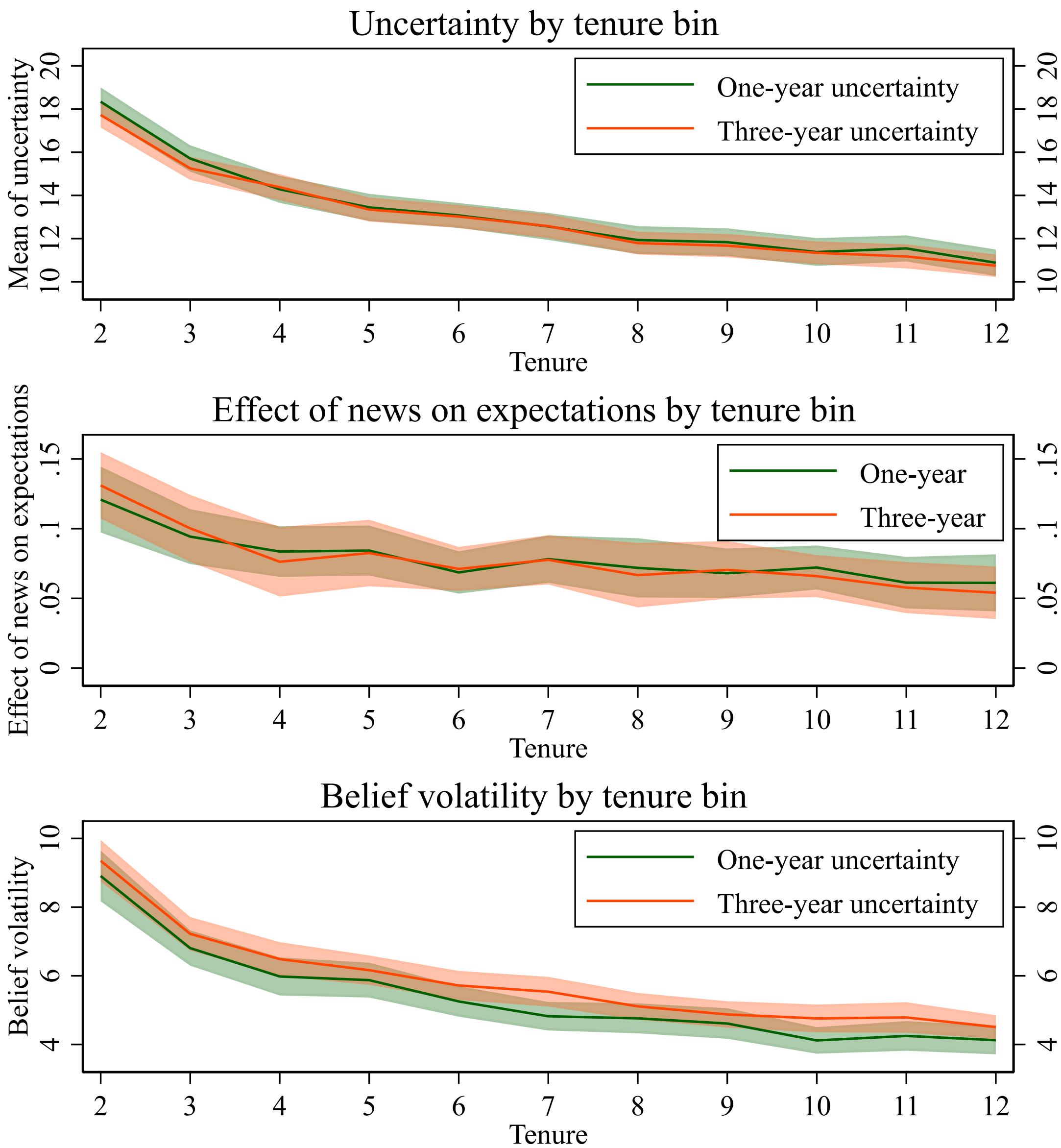
To illustrate the first stage, the top panel of figure 4 plots average uncertainty (the variance, $u_{i,t}$) at both the one- and three-year horizons by tenure in the survey. There is a clear negative relationship, which is statistically well estimated. Much of the decline is in the first few months of tenure, but even between months 7 and 12 there is a decline in average uncertainty of 14 percent.

For most of the analysis, we use a linear first stage. Table 1 reports estimation results for that (i.e. equation (12)). For all results here and in the rest of the paper, standard errors are heteroskedasticity robust with two-way clustering by person and time. The top panel reports results for one-year expectations and the bottom panel three-year expectations. The first two columns in each panel report results for regressions of uncertainty on tenure, restricting to either tenure>1 month or tenure>2 months, which corresponds to the two different subsamples that are used depending on the lag used for expectations in calculating inflation surprises.

In columns 1 and 2 in both panels, tenure is highly statistically significant, and the F-statistics are greater than 100, implying there are no concerns about weak instruments in this case.

[9] The 16 permutations are from the cross of four binary choices: (1) PCE or CPI; (2) headline or core; (3) one- or three-year horizon; (4) point forecast or distributional mean.

Figure 4: First stage and reduced-form relationships



Note: In each panel, the lines are point estimates and the shaded regions 95-percent confidence bands. Standard errors are two-way clustered by person and time. The top panel reports coefficients from a regression of uncertainty (each agent's variance) on tenure indicators. The middle panel is from a regression of the change in each agent's conditional mean on inflation news interacted with tenure indicators. The bottom panel is from a regression of the squared change in expectations on tenure indicators.

Columns 3 and 4 control for inflation surprises in addition to tenure. They differ depending on whether inflation surprises are measured based on the distributional mean or point forecast – with the mean, again, the independent variable is entered with two lags since it appears on the right-hand side in the baseline second stage. In columns 3–4 the t-statistic on tenure shrinks somewhat, but not enough to raise any weak identification concerns. Looking at the coefficients themselves, the inflation surprises absorb some of the effect of tenure because expectations themselves – which are part of the inflation surprise calculation – are also correlated with tenure.

Table 1: First stage regressions

(a) First Stage, 1 year inflation

	(1)	(2)	(3)	(4)
Dep var:	Unc.	Unc.	Unc.	Unc.
Tenure	−0.377***	−0.264***	−0.183***	−0.195***
	[−0.421, −0.333]	[−0.310, −0.218]	[−0.221, −0.145]	[−0.241, −0.149]
Infl. Surpr.			−0.602***	0.177***
			[−0.757, −0.447]	[0.078, 0.276]
Type of infl. surpr.			Point	Mean
F-statistic	279.182	124.769	87.305	38.462
R^2	0.004	0.002	0.029	0.003
Observations	116,720	104,595	103,888	93,460

(b) First Stage, 3 year inflation

	(1)	(2)	(3)	(4)
Dep var:	Unc.	Unc.	Unc.	Unc.
Tenure	−0.393***	−0.294***	−0.183***	−0.231***
	[−0.438, −0.348]	[−0.341, −0.247]	[−0.223, −0.143]	[−0.277, −0.185]
Infl. Surpr.			−0.601***	0.271***
			[−0.755, −0.447]	[0.176, 0.366]
Type of infl. surpr.			Point	Mean
F-statistic	294.550	151.649	89.453	59.977
R^2	0.004	0.002	0.026	0.005
Observations	117,019	104,827	101,438	92,765

Note: The table reports results of regressions of uncertainty on tenure and inflation surprises, with 1-year inflation expectations in panel (a) and 3-year expectations in panel (b). The first column restricts the sample to those with tenure above 1, the second to those with tenure above 2. Columns (3) and (4) use CPI headline inflation surprises constructed from the point forecast (lagged one month) and the bin mean (lagged two months), respectively. The regressions correspond to the first stage of the control-function estimation approach. 95% confidence intervals based on robust standard errors allowing for two-way clustering by person and time are reported in brackets. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.1 in the appendix reports results for the 2SLS first stage, which is a regression of the inflation surprise multiplied by uncertainty on the surprise and tenure times the surprise (i.e. it is the CF first stage with all variables interacted with the inflation surprise). While it is true that the F-statistics are large in all cases, they are driven by the presence of the inflation surprise on the right-hand side. The robust partial F-statistic on the excluded instrument – $\hat{\pi}_{i,t}z_{i,t}$ – is equal here to the squared t-statistic, which is 38.1 and 15.3 in the two columns of panel (a), and 18.8 and 20 for panel (b). The Montiel-Olea–Pflueger (2013) cutoff for this case is no smaller than 12, indicating that 2SLS is close to weakly identified here, demonstrating an additional advantage of the CF approach (see Wooldridge (2015)).

3.5 Reduced-form estimates

We first examine results without using an instrument. Table 2 reports results from regressions of $\Delta m_{i,t}$ on $\hat{\pi}_{i,t}$ and $\hat{\pi}_{i,t}u_{i,t}$ for the two inflation surprise measures (relative to the point forecast and distributional mean). Panel (a) reports results for 1-year expectations and panel (b) for 3-year expectations. Columns (1)–(2) use contemporaneous uncertainty and columns (3)–(4) use lagged uncertainty. As one would expect, positive inflation surprises are associated with increases in future expectations. However, the interaction of the surprise with uncertainty is, with a single exception, estimated to be *negative*, the opposite sign to what Bayesian updating predicts if the cross-sectional variation in uncertainty is exogenous. That is consistent with uncertainty being related to attention or to realized inflation.

In the regression, there are two sources of endogeneity: that uncertainty is driven by the same signals as expectations, and that uncertainty is driven by signal precision. A way to distinguish them is to lag uncertainty in the regression, which should avoid the problem of uncertainty responding to the signals that drive $m_{i,t}$. Columns (3)–(4) do that and show that lagged uncertainty also does not produce the positive interaction coefficient implied by the model. In the context of the model, that suggests that the source of the zero or negative coefficient is that the same people with high uncertainty also have the lowest attention (and hence lowest $\sigma_{i,t}^{-2}$). That is, some people pay a lot of attention to inflation, making them fairly confident in their forecasts. Others don't pay much attention and are much less confident. Those two effects offset each other so that on net there is no positive relationship between uncertainty and responsiveness. That is why we need an instrument: as a source of variation in uncertainty that is uncorrelated with attention in the cross-section.

3.5.1 News regressions

Columns (1) and (2) of table 3 report results for reduced-form regressions of $\Delta m_{i,t}$ on $\hat{\pi}_{i,t}$ and $\hat{\pi}_{i,t}z_{i,t}$. That is, compared to table 2, uncertainty is replaced with tenure. Since tenure is negatively associated with uncertainty, if it is also exogenous to attention in the cross-section the model predicts that the coefficient on the interaction should be negative. That is in fact what we observe across all the regressions in table 3.

Table 2: OLS reduced-form regressions of $\Delta m_{i,t}$ on inflation surprise and surprise $\times$ uncertainty

(a) 1 year inflation

	Dep. var.: $\Delta m_{i,t}$			
	Contemp. uncertainty		Lagged uncertainty	
	(1)	(2)	(3)	(4)
Unc. $\times$ Infl. Surpr.	-0.00124^{***}	-0.00075^{***}	-0.00062^{***}	0.00037^{***}
	[-0.00153, -0.00095]	[-0.00102, -0.00048]	[-0.00091, -0.00033]	[0.00010, 0.00064]
Infl. Surpr.	0.09405^{***}	0.03716^{***}	0.08756^{***}	0.02471^{***}
	[0.07676, 0.11134]	[0.02665, 0.04767]	[0.06992, 0.10520]	[0.01451, 0.03491]
Type of infl. surpr.	Point	Mean	Point	Mean
R^2	0.029	0.004	0.027	0.004
Obs.	103,888	93,460	103,888	93,460

(b) 3 year inflation

	Dep. var.: $\Delta m_{i,t}$			
	Contemp. uncertainty		Lagged uncertainty	
	(1)	(2)	(3)	(4)
Unc. $\times$ Infl. Surpr.	-0.00076^{***}	-0.00056^{***}	-0.00050^{***}	0.00017
	[-0.00100, -0.00052]	[-0.00080, -0.00032]	[-0.00073, -0.00027]	[-0.00008, 0.00042]
Infl. Surpr.	0.05313^{***}	0.02524^{***}	0.05045^{***}	0.01715^{***}
	[0.04130, 0.06496]	[0.01738, 0.03310]	[0.03865, 0.06225]	[0.00944, 0.02486]
Type of infl. surpr.	Point	Mean	Point	Mean
R^2	0.008	0.002	0.008	0.001
Obs.	101,438	92,765	101,438	92,765

Note: OLS regressions of changes in bin-mean expectations $\Delta m_{i,t}$ on the inflation surprise $\hat{\pi}_{i,t}$ and on the interaction of the surprise with subjective uncertainty $u_{i,t}\hat{\pi}_{i,t}$. Columns (1)–(2) use contemporaneous uncertainty $u_{i,t}$; columns (3)–(4) use lagged uncertainty $u_{i,t-1}$. Within each pair, the first column constructs the inflation surprise from the respondent's point forecast (lagged one month) and the second from the distributional mean (lagged two months). Panel (a) reports results for 1-year expectations and panel (b) for 3-year expectations. 95% confidence intervals based on robust standard errors allowing for two-way clustering by respondent and time are reported in brackets. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

To get a sense of magnitudes, the average sensitivity of one-year expectations to news at $tenure = 2$ in columns (1) and (2) is 0.10 and 0.04, compared to 0.06 and 0.02 at 12 months respectively for the two specifications. A person at the end of their time in the sample is therefore on average only half as responsive to inflation surprises as at the beginning. The sensitivity of three-year expectations also declines by about half. Table A.2 shows that similar results hold across the range of permutations of the specification.

The middle panel of figure 4 plots estimates from regressions of $\Delta m_{i,t}$ on tenure dummies multiplied by $\hat{\pi}_{i,t}$, yielding estimates of sensitivity to surprises at each level of tenure without imposing linearity in the interaction. The negative relationship is clearly apparent. Again, the estimates imply that sensitivity to inflation surprises falls by about half for agents at the end of their tenure compared to the beginning and that is consistent with the result from the top panel that uncertainty also falls by about half over the same period.

Note, importantly, that the decline by a factor of one half in both uncertainty and responsiveness is consistent with the model's proportionality prediction. Similar results will

Table 3: Reduced form regressions

(a) 1 year inflation

	Dep. var.: Δm			Dep. var.: $(\Delta m)^2$
	(1)	(2)		(3)
Tenure × Infl. Surpr.	-0.0046^{***}	-0.0021^{***}	Tenure	-0.40^{***}
	[−0.0057, −0.0035]	[−0.0032, −0.0010]		[−0.44, −0.36]
Infl. Surpr.	0.1112^{***}	0.0445^{***}		
	[0.0931, 0.1293]	[0.0326, 0.0564]		
Type of infl. surpr.	Point	Mean		
R^2	0.0270	0.0040	R^2	0.01
Obs.	103,888	93,460	Obs.	118,141

(b) 3 year inflation

	Dep. var.: Δm			Dep. var.: $(\Delta m)^2$
	(1)	(2)		(3)
Tenure × Infl. Surpr.	-0.0025^{***}	-0.0007	Tenure	-0.40^{***}
	[−0.0036, −0.0014]	[−0.0018, 0.0004]		[−0.44, −0.36]
Infl. Surpr.	0.0611^{***}	0.0244^{***}		
	[0.0472, 0.0750]	[0.0126, 0.0362]		
Type of infl. surpr.	Point	Mean		
R^2	0.0070	0.0010	R^2	0.01
Obs.	101,438	92,765	Obs.	118,620

Note: The left side of the table shows reduced-form regressions of changes in bin mean expectations ($\Delta m_{i,t}$) on tenure interacted with inflation surprises, as well as the level of the inflation surprise. The right side regresses the squared changes in expectations $(\Delta m_{i,t})^2$ on tenure. Columns (1)–(2) vary the construction of the inflation surprise using either the point forecast (lagged by one month) or the bin mean (lagged by two months). 95% confidence intervals based on robust standard errors allowing for two-way clustering by respondent and time are reported in brackets. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

hold throughout the empirical analysis.

Kim and Binder (2023) show that in addition to uncertainty declining with tenure, average inflation expectations do, also. That should cause the average of $\Delta m_{i,t}$ to be negative. If the rate of decline changes with tenure, then tenure would naturally also be included as a control in these regressions. Table A.3 in the appendix reports results for regressions analogous to those in table 3 but including tenure as an additional control. The coefficient on the interaction $\hat{\pi}_{i,t} z_{i,t}$ is essentially unaffected. The estimated coefficient on tenure is positive and the constant in the regression negative, consistent with average expectations declining, but at a slowing rate, with tenure.

3.5.2 Volatility in expectations

The third column of table 3 reports results from regressions of $(\Delta m_{i,t})^2$ on tenure. The coefficient is again significantly negative, implying that as tenure increases agents are less responsive to information overall (under the model that is because their expectations become more stable). The bottom panel of figure 4 plots the average of $(\Delta m_{i,t})^2$ by tenure bins. The negative relationship holds across the entire range of tenure and is not driven by any particular bin. Note also that the decline is very similar in shape to the decline in uncertainty itself. Again, consistent with the proportionality prediction of the model, the approximately 50% proportional decline in belief volatility matches the proportional decline in uncertainty across tenure bins.

3.6 Second-stage estimates

3.6.1 News regressions

Table 4 reports results from the second stage of the control function estimator – equation (16) – where the conditional expectations are, in the benchmark results, approximated as linear functions of $\eta_{i,t}$ and $\hat{\pi}_{i,t}$.[10] The first row reports the mean sensitivity of expectations to $u_{i,t}\hat{\pi}_{i,t}$ (i.e. at the average value of the various interactions) which, under the model, represents the cross-sectional mean of the precision of the signal each agent gets from inflation, $\sigma_{i,t}^{-2}$. (The full specification of the regression is reported in appendix table A.4.) That coefficient ranges between 0.0032 and 0.0229, depending on the specification, and it is statistically well estimated (insignificant only in column (4)). The point estimates imply that the standard deviation in the noise in agents' signals about future inflation from current realized inflation is between 7 and 18 percent. Recall from the model that that noise combines both the gap between realized inflation and its persistent component and also the difference between what agents observe and actual realized inflation.

The second row of table 4 reports the mean sensitivity of expectations to $\eta_{i,t}\hat{\pi}_{i,t}$, which measures how the sensitivity of expectations to news depends on the *endogenous* part of uncertainty – the part not driven by tenure. The fact that the coefficients are negative is consistent with the idea that agents with high uncertainty are probably people who pay less attention to inflation on average, so that $\sigma_{i,t}^{-2}$ is low, with the result that the net relationship between sensitivity to inflation and $\eta_{i,t}$ need not be positive.

[10]Standard errors here are constructed by putting the first and second stage regressions into a single GMM problem so that the standard errors reported in table 4 account for uncertainty in the first stage. As above, standard errors are two-way clustered by person and time.

Table 4: Second Stage Regression

	1 year inflation		3 year inflation	
	(1)	(2)	(3)	(4)
Unc. $\times$ Infl. Surpr., AME	0.0229***	0.0114***	0.0120***	0.0032
	[0.0163, 0.0295]	[0.0049, 0.0179]	[0.0057, 0.0183]	[−0.0019, 0.0083]
$\eta_{i,t}$ $\times$ Infl. Surpr., AME	−0.0245***	−0.0127***	−0.0134***	−0.0046*
	[−0.0311, −0.0179]	[−0.0191, −0.0063]	[−0.0198, −0.0070]	[−0.0098, 0.0006]
Other controls	Y	Y	Y	Y
Type of infl. surpr.	Point	Mean	Point	Mean
Observations	103,888	93,460	101,438	92,765

Note: Results from second-stage regressions of $\Delta m_{i,t}$ (measured based on the bin mean) on interactions of uncertainty (Unc. = $u_{i,t}$), inflation surprise ($\hat{\pi}_{i,t}$), and control function ($\eta_{i,t}$). The regression includes the terms obtained from the Kroneker product of (inflation surprise; uncertainty$\times$inflation surprise) with (a constant, the control function component, and inflation surprise). All the coefficients of this regression are reported in table A.4. AME denotes average marginal effect. This table reports the mean sensitivity of expectations to $u_{i,t}\hat{\pi}_{i,t}$ in the first row and the mean sensitivity of expectations to $\eta_{i,t}\hat{\pi}_{i,t}$ (both computed at the average values of the various interactions). Columns (1)–(2) use different versions of the inflation surprise—measured based on the point forecast (Point) or the lagged bin mean (Mean)—for the 1-year inflation expectations. Columns (3)–(4) report the analogous specifications for the 3-year inflation expectations. 95% confidence intervals in brackets account for sampling uncertainty in the first stage and allow for clustering by both respondent and time. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.5 reports results from the other 12 permutations of the specification choices (CPI vs PCE, etc.). While there is variation in the coefficient estimates for $u_{i,t}\hat{\pi}_{i,t}$, they are broadly consistent with each other and uniformly positive and significant. Table A.6 replicates table 4 but using agents' point forecast for inflation on the left-hand side instead of their implied mean. Finally, table A.7 reports results from second-stage regressions including richer controls – most importantly, higher powers of various variables (recalling the sieve interpretation of the regression) – and finds similar results.

For the sake of completeness, table A.8 in the appendix reports the 2SLS second stage estimates. The coefficients are similar to those obtained via the CF method, as are the standard errors, suggesting that in practice both the cross-sectional heterogeneity and first-stage concerns with 2SLS may not be particularly important.

3.6.2 Volatility in expectations

Table 5 reports results for the second-stage regression of $(\Delta m_{i,t})^2$ on uncertainty – equation (17). The estimated sensitivity to $u_{i,t}^2$ is, under the conditions discussed above, again an estimate of $E\left[\sigma_{i,t}^{-2}\right]$, where here that represents the precision of the total signal that the agents receive in period t. Intuitively, the idea behind the regression is that, all else equal, when an agent's uncertainty is higher, the variance of the change in their expectations should also be higher, in proportion to the precision of their signals.

Table 5: Second Stage Regression for $(\Delta m)^2$

	1-year inflation		3-year inflation	
	(1)	(2)	(3)	(4)
$(Unc.)^2$, AME	0.1768***	0.0942***	0.2483***	0.0862***
	[0.1290, 0.2246]	[0.0805, 0.1079]	[0.1888, 0.3078]	[0.0740, 0.0984]
$\eta_{i,t}^2$, AME	0.0216**	0.0121***	0.0543***	0.0207***
	[0.0029, 0.0403]	[0.0070, 0.0172]	[0.0323, 0.0763]	[0.0162, 0.0252]
Type of inflation expectation	Point	Mean	Point	Mean
Observations	103,371	118,141	102,113	118,620

Note: Results from second-stage regressions of $(\Delta m_{i,t})^2$ on interactions of uncertainty and the control-function component. Uncertainty is $Unc_{i,t} = u_{i,t}$. The regression includes terms obtained from the Kronecker product $(1,\, Unc_{i,t}^2) \otimes (1,\, \eta_{i,t},\, \eta_{i,t}^2,\, \eta_{i,t}^{\text{inv}})$, where $\eta_{i,t}$ is demeaned and $\eta_{i,t}^{\text{inv}} = (\eta_{i,t} - \min(\eta_{i,t}) + 10)^{-1}$. All coefficients from the second-stage regression are reported in the appendix. This table reports the AME (average marginal effect) of $Unc_{i,t}^2$ and the AME of the control-function component, both evaluated at the sample-average values of the interaction terms. Columns (1)–(2) use 1-year inflation expectations and columns (3)–(4) use 3-year inflation expectations; within each horizon, the dependent variable is constructed using point expectations (Point) or lagged bin means (Mean). 95% confidence intervals in brackets account for sampling uncertainty in the first stage and allow for clustering by both respondent and time. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The estimated values of $E\left[\sigma_{i,t}^{-2}\right]$ are significantly larger in table 5 than in table 4. That is exactly what we should expect – agents receive many signals about future inflation beyond realized current inflation. The results in table 4 indicated that the signal agents got from inflation had an error with effective standard deviation of 7–18%. Table 5 implies that the standard deviation of the effective error in their *overall* signal is between 2% and 3.4%.

The second row of table 5 reports the sensitivity to $\eta_{i,t}^2$. In this case, in contrast to table 4, the coefficients are positive. The intuition for that result is as follows. Suppose some agents just pay less attention to the economy and know less about inflation, thus reporting distributions that tend to be wider. In table 4, the negative coefficient on $\eta_{i,t}\hat{\pi}_{i,t}$ indicates those agents' inflation expectations are less responsive to realized inflation than average. At the same time, it is plausible that those agents also report expectations that vary somewhat erratically – given that they do not pay much attention, there is not much reason to expect any real stability in their reported beliefs. That is why the coefficient in the second row of table 5 is positive.

It should be noted that that suggests a potential source of bias in the volatility regressions. Some of the variation in Δm_t may just be noise – e.g. due to difficulty people have in translating expectations into specific numbers. If that noise declines with tenure, that will create an upward bias in the estimate of $E\left[\sigma_{i,t}^{-2}\right]$. The regressions for the trend-plus-noise model above are not exposed to this issue because any noise in expectations is absorbed into the residual.

4 Time-series analysis

While the panel analysis is useful for directly testing models of beliefs, in practice policymakers typically focus on the cross-sectional mean or median of beliefs. Additionally, what is relevant for policymakers is most likely *forecasting* the sensitivity of beliefs to signals, as opposed to the contemporaneous relationships the analysis in the previous section tests. This section therefore examines whether current reported uncertainty in the SCE predicts either future sensitivity of average expectations to inflation or the magnitude of squared changes in beliefs. It also examines whether the aggregate relationship is consistent with the structural estimate of the average of $\sigma_{i,t}^{-2}$.

The theoretical analysis stressed the endogeneity of uncertainty to signal precision. Taking a cross-sectional average helps alleviate that concern because while it seems very likely that there are people with high uncertainty because they pay little attention to inflation, it seems less likely that variation over time in average uncertainty is driven by variation over time in the precision of signals agents observe. It is certainly not *impossible*, just not as prominent an issue. Furthermore, in this case we are not so much trying to test the model as simply ask whether uncertainty is useful for predicting sensitivity to shocks, which, again, is a forecasting problem and not a causal identification problem.

To align with typical aggregate time series, we first collapse the panel structure into cross-sectional medians in each period. We then define inflation news as inflation in month t minus the cross-sectional median of 12-month expected inflation (from the distributional mean) in month $t-2$. The expectation is lagged by two months because the change in inflation, which is on the left-hand side of the regression, involves expected inflation in month $t-1$.

Table 6 reports results from four regressions. The first two are of the change in median expectations at the one- and three-year horizons on inflation news, both alone and interacted with the cross-sectional median of uncertainty at date $t-1$ (u_{t-1}).

In the sample, the standard deviations of one- and three-year uncertainty are 1.4 and 1.0, respectively. In the first column, the coefficient on the interaction of news with uncertainty is 0.009, while the coefficient on news alone is actually slightly negative and not statistically distinguishable from zero. Both of those are consistent with the model. The regression implies that if, hypothetically, people had zero uncertainty about future inflation, their expectations would be completely unresponsive to realized inflation. Additionally, the value

Table 6: Aggregate Analysis

Dep. var.:	(1) Δm (1yr)	(2) Δm (3yr)	(3) $(\Delta m)^2$ (1yr)	(4) $(\Delta m)^2$ (3yr)
Infl. Surprise	−0.00563	−0.00117		
	[−0.02815, 0.01689]	[−0.01755, 0.01521]		
Lagged unc.	−0.01638	−0.01846		
	[−0.04786, 0.01510]	[−0.04673, 0.00981]		
Surprise × Lagged unc.	0.00870***	0.00456**		
	[0.00371, 0.01369]	[0.00028, 0.00884]		
Lagged (unc.)^2			0.00372***	0.00197***
			[0.00229, 0.00515]	[0.00093, 0.00301]
Constant	0.05455	0.05497	−0.00307	0.00018
	[−0.01769, 0.12679]	[−0.01883, 0.12877]	[−0.01464, 0.00850]	[−0.00864, 0.00900]
F-statistic	8.62	13.24	25.89	13.82
R^2	0.223	0.120	0.262	0.138
Observations	140	140	141	141

Note: Column (1) reports results of a time-series regression of the change in aggregate inflation expectations (cross-sectional median of individual expectations) on CPI inflation surprise, computed as realized inflation minus two-month-lagged cross-sectional median expectation, lagged aggregate uncertainty (cross-sectional median of individual uncertainty), and their interaction. Column (2) repeats the exercise using 3-year inflation expectations and uncertainty. Column (3) regresses squared changes in 1-year aggregate inflation expectations on lagged squared uncertainty. Column (4) repeats the exercise using 3-year changes in inflation expectation and uncertainty. 95% confidence intervals based on Newey-West standard errors with 12 lags are reported in brackets. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

of the coefficient on uncertainty is consistent with the structural estimate of mean signal precision from table 4. That is, the variation in the sensitivity of average expectations that is related to variation in average uncertainty is approximately of the magnitude that would be expected from the variation in sensitivity observed across months of tenure within respondents.

The results are similar for three-year expectations. Again, the coefficient on the inflation surprise by itself is statistically indistinguishable from zero, and the coefficient on the interaction with uncertainty is 0.005, a similar magnitude to the micro estimate.

The left-hand panel of figure 5 plots the fitted sensitivity of one- and three-year expectations from the two regressions. There is a striking degree of variation. At its peak in 2022, the sensitivity of one-year expectations is estimated to be 0.06. Exactly when inflation was highest, expectations were most sensitive to inflation surprises. That sensitivity was higher than the value at the beginning of 2020 by a factor of four. Furthermore, at the end of the sample in 2025, sensitivity remains nearly twice as high as it was in 2020.

Similar results, though at lower overall levels, hold for three-year expectations. Their sensitivity to inflation surprises peaked at 0.04, and they are also now about twice their pre-covid values.

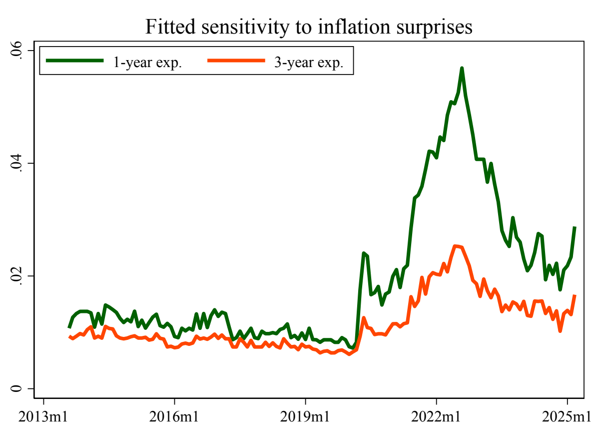
The second pair of columns in table 6 report results from regressions of squared changes in median expectations on lagged squared uncertainty. As in the first two columns, the constants in the regressions are not statistically significant, consistent with the model.

The coefficients on uncertainty are again highly statistically significant, but in this case, unlike the first two columns, the estimated coefficients do not line up well with those from the person-level estimates. That is consistent with the potential bias discussed above if there is noise in individuals' reports of their expectations.

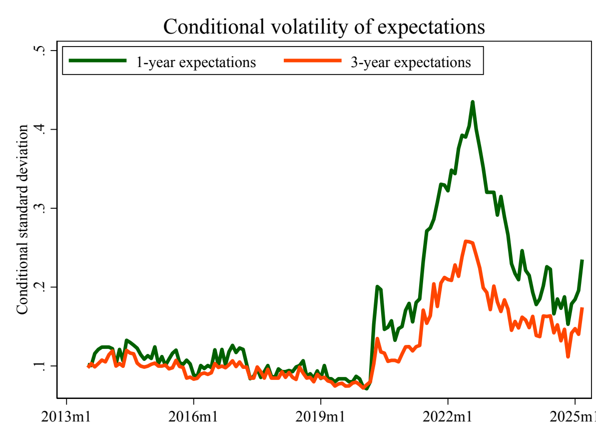
Again, the coefficients on squared uncertainty in these columns should be estimates of average signal precision. They are smaller than those in the first two columns, but of broadly similar magnitude, and their confidence bands overlap substantially.

To see the effects of time-series variation in uncertainty on the predicted volatility, the right-hand panel of figure 5 plots the square roots of the fitted values from the regressions from columns (3) and (4) of table 6. These values represent conditional standard deviations for one- and three-year median inflation expectations under the assumption that the expectations are martingales. Those standard deviations were both equal to about 0.07 at the beginning of 2020. They rose by a factor of 4–6 in 2022 and by 2025 remained more than twice as high as pre-covid.

Figure 5: Aggregate time-series regressions: sensitivity and conditional volatility



(a) Fitted sensitivity to inflation surprises



(b) Fitted conditional volatility of expectations

Note: Panel (a) plots fitted values of the time-varying sensitivity of changes in median inflation expectations to inflation surprises for the 1-year and 3-year horizons. Panel (b) plots the square roots of fitted values from the regressions of squared changes in median expectations on lagged squared uncertainty. Standard errors in the underlying regressions are Newey–West with 12 lags.

5 Conclusion

This paper studies the anchoring of inflation expectations. Starting from the observation that the response function of expectations to news is fundamentally unobservable, its key insight is that under the assumptions that agents are Bayesian and that they observe Gaussian signals, the response function can be recovered from knowledge of agents' posterior distributions. And in fact the recovery is not even particularly complicated: the derivatives of the response function are simply the posterior cumulants.

Obviously the structural assumptions are strong. That is somewhat inevitable since the goal is to measure something unobservable. Ultimately the empirical results have to speak for themselves: does an agent's reported uncertainty actually have any predictive power for the sensitivity of their beliefs to news? Across a variety of measures – contemporaneous and forward-looking, time-series and cross-sectional – the answer is broadly yes.

The results imply that the period of high inflation in 2021 and 2022 has had, so far, lasting effects on the strength of the expectational anchor. Agents' uncertainty about future inflation rose as inflation did, and subsequently declined, but not nearly to the extent that inflation itself did. By the beginning of 2025 inflation was approaching its pre-2020 levels, but inflation uncertainty was still about three times higher. Those facts show how the results in this paper can be used going forward. They provide a real-time quantitative measure of how well inflation expectations are anchored. More generally, they show how to measure the sensitivity of expectations to news in arbitrary settings.

References

Ameriks, John, Andrew Caplin, Steven Laufer, and Stijn Van Nieuwerburgh, "The joy of giving or assisted living? Using strategic surveys to separate public care aversion from bequest motives," *The journal of finance*, 2011, *66* (2), 519–561.

Armantier, Olivier, Argia Sbordone, Giorgio Topa, Wilbert van der Klaauw, and John C. Williams, "A New Approach to Assess Inflation Expectations Anchoring Using Strategic Surveys," *Journal of Monetary Economics*, 2022, *129*, S82–S101.

_ , Scott Nelson, Giorgio Topa, Wilbert Van der Klaauw, and Basit Zafar, "The price is right: Updating inflation expectations in a randomized price information experiment," *Review of Economics and Statistics*, 2016, *98* (3), 503–523.

— , **Wändi Bruine de Bruin, Giorgio Topa, Wilbert Van Der Klaauw, and Basit Zafar**, "Inflation expectations and behavior: Do survey respondents act on their beliefs?," *International Economic Review*, 2015, *56* (2), 505–536.

Ball, Laurence M. and Sandeep Mazumder, "Inflation Dynamics and the Great Recession," *Brookings Papers on Economic Activity*, 2011, pp. 337–381.

Binder, Carola Conces, Jeffrey R. Campbell, and Jane M. Ryngaert, "Consumer Inflation Expectations: Daily Dynamics," 2024.

Bordalo, Pedro, Nicola Gennaioli, Rafael La Porta, and Andrei Shleifer, "Diagnostic expectations and stock returns," *The Journal of Finance*, 2019, *74* (6), 2839–2874.

Carvalho, Carlos, Stefano Eusepi, Emanuel Moench, and Bruce Preston, "Anchored inflation expectations," *American Economic Journal: Macroeconomics*, 2023, *15* (1), 1–47.

Cavallo, Alberto, Guillermo Cruces, and Ricardo Perez-Truglia, "Inflation expectations, learning, and supermarket prices: Evidence from survey experiments," *American Economic Journal: Macroeconomics*, 2017, *9* (3), 1–35.

Coibion, Olivier and Yuriy Gorodnichenko, "Information rigidity and the expectations formation process: A simple framework and new facts," *American Economic Review*, 2015, *105* (8), 2644–2678.

— **and** — , "Inflation Expectations and Monetary Policy?What Have We Learned and To What End?," 2025. Working paper.

D'Acunto, Francesco and Michael Weber, "Why Survey-Based Subjective Expectations areMeaningful and Important," *Annual Review of Economics*, 2024.

Daniel, Kent, David Hirshleifer, and Avanidhar Subrahmanyam, "Investor Psychology and Security Market Under- and Overreactions," *Journal of Finance*, 1998, *53* (6), 1839–1885.

DeMarzo, Peter M, Dimitri Vayanos, and Jeffrey Zwiebel, "Persuasion bias, social influence, and unidimensional opinions," *The Quarterly journal of economics*, 2003, *118* (3), 909–968.

Dew-Becker, Ian, Stefano Giglio, and Pooya Molavi, "Learning and the emergence of nonlinearity in financial markets," 2026. Working paper.

Dytso, Alex, H Vincent Poor, and Shlomo Shamai Shitz, "Conditional mean estimation in Gaussian noise: A meta derivative identity with applications," *IEEE Transactions on Information Theory*, 2022, *69* (3), 1883–1898.

Farmer, Leland E, Emi Nakamura, and Jón Steinsson, "Learning about the long run," *Journal of Political Economy*, 2024, *132* (10), 000–000.

Fleckenstein, Matthias, Francis A Longstaff, and Hanno Lustig, "Deflation risk," *The Review of Financial Studies*, 2017, *30* (8), 2719–2760.

Hall, Peter and Christopher C Heyde, *Martingale limit theory and its application*, Academic press, 2014.

Hilscher, Jens, Alon Raviv, and Ricardo Reis, "How likely is an inflation disaster?," *The Review of Financial Studies*, 2025, p. hhaf058.

Kim, Gwangmin and Carola Binder, "Learning-through-survey in inflation expectations," *American Economic Journal: Macroeconomics*, 2023, *15* (2), 254–278.

Kitsul, Yuriy and Jonathan H Wright, "The economics of options-implied inflation probability density functions," *Journal of Financial Economics*, 2013, *110* (3), 696–711.

Kumar, Saten, Hassan Afrouzi, Olivier Coibion, and Yuriy Gorodnichenko, "Inflation targeting does not anchor inflation expectations: Evidence from firms in New Zealand," *Brookings Papers on Economic Activity*, 2015, *2015* (2), 151–225.

Lamla, Michael J and Dmitri V Vinogradov, "Central bank announcements: Big news for little people?," *Journal of Monetary Economics*, 2019, *108*, 21–38.

Malmendier, Ulrike and Stefan Nagel, "Learning from inflation experiences," *The Quarterly Journal of Economics*, 2016, *131* (1), 53–87.

Masten, Matthew A and Alexander Torgovitsky, "Identification of instrumental variable correlated random coefficients models," *Review of Economics and Statistics*, 2016, *98* (5), 1001–1005.

Mertens, Thomas M and John C Williams, "What to expect from the lower bound on interest rates: Evidence from derivatives prices," *American Economic Review*, 2021, *111* (8), 2473–2505.

Molavi, Pooya, "Misspecified Bayesianism," 2025. Working paper.

Newey, Whitney and Sami Stouli, "Control variables, discrete instruments, and identification of structural functions," *Journal of Econometrics*, 2021, *222* (1), 73–88.

Newey, Whitney K, "Convergence rates and asymptotic normality for series estimators," *Journal of econometrics*, 1997, *79* (1), 147–168.

Reis, Ricardo, "Why did inflation rise and fall in 2021-24?Channels and evidence from expectations," 2025. Working paper.

Roth, Christopher and Johannes Wohlfart, "How do expectations about the macroeconomy affect personal expectations and behavior?," *Review of Economics and Statistics*, 2020, *102* (4), 731–748.

Weber, Michael, Bernardo Candia, Hassan Afrouzi, Tiziano Ropele, Rodrigo Lluberas, Serafin Frache, Brent Meyer, Saten Kumar, Yuriy Gorodnichenko, Dimitris Georgarakos et al., "Tell Me Something I Don't Already Know: Learning in Low-and High-Inflation Settings," *Econometrica*, 2025, *93* (1), 229–264.

Wooldridge, Jeffrey M, "Control function methods in applied econometrics," 2015, *50* (2), 420–445.

Table A.1: First stage regressions, 2SLS approach

(a) First stage, 1 year inflation

	(1)	(2)
Dep var:	Unc. × Surp.	Unc. × Surp.
Tenure × Surpr.	−0.325***	−0.183***
	[−0.428, −0.222]	[−0.275, −0.091]
Infl. Surpr.	13.817***	12.228***
	[12.352, 15.282]	[11.243, 13.213]
Type of infl. surpr.	Point	Mean
F-statistic	188.379	319.883
R^2	0.217	0.229
Observations	103,888	93,460

(b) First stage, 3 year inflation

	(1)	(2)
Dep var:	Unc. × Surp.	Unc. × Surp.
Tenure × Surpr.	−0.266***	−0.226***
	[−0.386, −0.146]	[−0.325, −0.127]
Infl. Surpr.	13.748***	12.631***
	[12.340, 15.156]	[11.757, 13.505]
Type of infl. surpr.	Point	Mean
F-statistic	221.682	460.780
R^2	0.215	0.231
Observations	101,438	92,765

Note: The table (with panel (a) focusing on 1-year inflation and panel (b) on 3-year inflation) reports results of regressing the interaction of uncertainty and surprise on the interaction of tenure and surprise, as well as inflation surprise, corresponding to the first stage of the 2SLS approach. 95% confidence intervals based on robust standard errors allowing for two-way clustering by respondent and time are reported in brackets. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.2: Reduced Form Regression: Robustness

(a) 1 year inflation

	(1)	(2)	(3)	(4)	(5)	(6)
Dep var:	Δm	Δm	Δm	Δm	Δm	Δm
Tenure × Infl. Surpr.	−0.0061***	−0.0025***	−0.0056***	−0.0027***	−0.0064***	−0.0031***
	[−0.0074, −0.0048]	[−0.0040, −0.0010]	[−0.0068, −0.0044]	[−0.0040, −0.0014]	[−0.0077, −0.0051]	[−0.0046, −0.0016]
Infl. Surpr.	0.1510***	0.0488***	0.1419***	0.0555***	0.1650***	0.0570***
	[0.1334, 0.1686]	[0.0363, 0.0613]	[0.1272, 0.1566]	[0.0430, 0.0680]	[0.1515, 0.1785]	[0.0445, 0.0695]
CPI or PCE	CPI	CPI	PCE	PCE	PCE	PCE
Headline or core	C	C	H	H	C	C
Point or Mean	P	M	P	M	P	M
R^2	0.032	0.003	0.034	0.004	0.036	0.003
Observations	103,888	93,460	103,888	93,460	103,888	93,460

(b) 3 year inflation

	(1)	(2)	(3)	(4)	(5)	(6)
Dep var:	Δm	Δm	Δm	Δm	Δm	Δm
Tenure × Infl. Surpr.	−0.0039***	−0.0021***	−0.0031***	−0.0014**	−0.0040***	−0.0024***
	[−0.0053, −0.0025]	[−0.0036, −0.0006]	[−0.0044, −0.0018]	[−0.0028, −0.0000]	[−0.0055, −0.0025]	[−0.0040, −0.0008]
Infl. Surpr.	0.0887***	0.0383***	0.0800***	0.0356***	0.0964***	0.0441***
	[0.0739, 0.1035]	[0.0224, 0.0542]	[0.0664, 0.0936]	[0.0227, 0.0485]	[0.0829, 0.1099]	[0.0293, 0.0589]
CPI or PCE	CPI	CPI	PCE	PCE	PCE	PCE
Headline or core	C	C	H	H	C	C
Point or Mean	P	M	P	M	P	M
R^2	0.009	0.001	0.010	0.002	0.011	0.002
Observations	101,438	92,765	101,438	92,765	101,438	92,765

Note: Robustness table for the reduced form regressions reported in table 3. Across columns, inflation surprises vary along three dimensions: (i) CPI vs. PCE inflation, (ii) headline vs. core, and (iii) point estimate vs. mean of the belief distribution. 95% confidence intervals based on robust standard errors allowing for clustering by respondent and time are reported in brackets. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.3: Reduced Form Regression: Robustness, controlling for tenure

(a) 1 year inflation

	(1)	(2)	(3)	(4)	(5)	(6)
Dep var:	Δm	Δm	Δm	Δm	Δm	Δm
Tenure × Infl. Surpr.	−0.0053***	−0.0021***	−0.0049***	−0.0023***	−0.0059***	−0.0027***
	[−0.0068, −0.0038]	[−0.0035, −0.0007]	[−0.0063, −0.0035]	[−0.0036, −0.0010]	[−0.0075, −0.0043]	[−0.0042, −0.0012]
Infl. Surpr.	0.1456***	0.0456***	0.1373***	0.0527***	0.1617***	0.0538***
	[0.1271, 0.1641]	[0.0336, 0.0576]	[0.1211, 0.1535]	[0.0396, 0.0658]	[0.1463, 0.1771]	[0.0414, 0.0662]
Tenure	0.0067***	0.0051***	0.0055**	0.0038**	0.0033	0.0035**
	[0.0026, 0.0108]	[0.0021, 0.0081]	[0.0013, 0.0097]	[0.0005, 0.0071]	[−0.0009, 0.0075]	[0.0005, 0.0065]
CPI or PCE	CPI	CPI	PCE	PCE	PCE	PCE
Headline or core	C	C	H	H	C	C
Point or Mean	P	M	P	M	P	M
R^2	0.032	0.003	0.034	0.004	0.036	0.003
Observations	103,888	93,460	103,888	93,460	103,888	93,460

(b) 3 year inflation

	(1)	(2)	(3)	(4)	(5)	(6)
Dep var:	Δm	Δm	Δm	Δm	Δm	Δm
Tenure × Infl. Surpr.	−0.0033***	−0.0020**	−0.0024***	−0.0012	−0.0033***	−0.0023***
	[−0.0049, −0.0017]	[−0.0036, −0.0004]	[−0.0039, −0.0009]	[−0.0026, 0.0002]	[−0.0051, −0.0015]	[−0.0040, −0.0006]
Infl. Surpr.	0.0838***	0.0372***	0.0745***	0.0341***	0.0918***	0.0435***
	[0.0684, 0.0992]	[0.0210, 0.0534]	[0.0599, 0.0891]	[0.0208, 0.0474]	[0.0767, 0.1069]	[0.0281, 0.0589]
Tenure	0.0062***	0.0019	0.0065***	0.0020	0.0046**	0.0007
	[0.0024, 0.0100]	[−0.0015, 0.0053]	[0.0024, 0.0106]	[−0.0015, 0.0055]	[0.0004, 0.0088]	[−0.0026, 0.0040]
CPI or PCE	CPI	CPI	PCE	PCE	PCE	PCE
Headline or core	C	C	H	H	C	C
Point or Mean	P	M	P	M	P	M
R^2	0.009	0.001	0.010	0.002	0.011	0.002
Observations	101,438	92,765	101,438	92,765	101,438	92,765

Note: Same as table A.2, but adding tenure as control. 95% confidence intervals based on robust standard errors allowing for clustering by respondent and time are reported in brackets. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.4: Second Stage Regression: Coefficients

	1 year inflation		3 year inflation	
	(1)	(2)	(3)	(4)
Unc. × Infl. Surpr.	0.0230***	0.0114***	0.0120***	0.0032
	[0.0164, 0.0296]	[0.0049, 0.0179]	[0.0057, 0.0183]	[−0.0019, 0.0083]
Unc. × $\eta_{i,t}$ × Infl. Surpr.	0.0000160***	0.0000075**	0.0000166***	0.0000110***
	[0.0000090, 0.0000230]	[0.0000011, 0.0000139]	[0.0000109, 0.0000223]	[0.0000040, 0.0000180]
Unc. × Infl. Surpr.^{2}	0.0000411	−0.0000025	0.0000212	−0.0000076
	[−0.0000092, 0.0000914]	[−0.0000433, 0.0000383]	[−0.0000173, 0.0000597]	[−0.0000490, 0.0000338]
Infl. Surpr.	−0.0804***	−0.0667**	−0.0458**	−0.0105
	[−0.1242, −0.0366]	[−0.1214, −0.0120]	[−0.0909, −0.0007]	[−0.0552, 0.0342]
Infl. Surpr.^{2}	−0.0017	0.0007	−0.0011	0.0004
	[−0.0045, 0.0011]	[−0.0010, 0.0024]	[−0.0031, 0.0009]	[−0.0009, 0.0017]
$\eta_{i,t}$× Infl. Surpr.	−0.0247***	−0.0128***	−0.0135***	−0.0047*
	[−0.0313, −0.0181]	[−0.0192, −0.0064]	[−0.0199, −0.0071]	[−0.0099, 0.0005]
$\eta_{i,t}$	0.0069***	0.0033***	0.0057***	0.0031***
	[0.0050, 0.0088]	[0.0021, 0.0045]	[0.0043, 0.0071]	[0.0021, 0.0041]
Constant	0.0921***	0.0110	0.0424***	0.0028
	[0.0479, 0.1363]	[−0.0119, 0.0339]	[0.0141, 0.0707]	[−0.0175, 0.0231]
Type of infl. surpr.	Point	Mean	Point	Mean
Observations	103,888	93,460	101,438	92,765

Note: Table reports all the coefficients of the regression in table 4. 95% confidence intervals based on standard errors that account for sampling uncertainty in the first stage and allow for clustering by both respondent and time are reported in brackets. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.5: Second Stage Regression: Robustness

(a) 1-Year Inflation

	Dep. var.: $\Delta m_{i,t}$					
	(1)	(2)	(3)	(4)	(5)	(6)
Unc. $\times$ Infl. Surpr., AME	0.0287***	0.0135***	0.0266***	0.0144***	0.0294***	0.0158***
	[0.0204, 0.0370]	[0.0050, 0.0220]	[0.0193, 0.0339]	[0.0068, 0.0220]	[0.0212, 0.0376]	[0.0072, 0.0244]
$\eta_{i,t}$ $\times$ Infl. Surpr., AME	−0.0317***	−0.0157***	−0.0289***	−0.0164***	−0.0327***	−0.0184***
	[−0.0400, −0.0234]	[−0.0241, −0.0073]	[−0.0362, −0.0216]	[−0.0239, −0.0089]	[−0.0409, −0.0245]	[−0.0268, −0.0100]
Other controls	Y	Y	Y	Y	Y	Y
CPI or PCE	CPI	CPI	PCE	PCE	PCE	PCE
Headline or core	C	C	H	H	C	C
Point or Mean	P	M	P	M	P	M
Observations	103,888	93,460	103,888	93,460	103,888	93,460

(b) 3-Year Inflation

	Dep. var.: $\Delta m_{i,t}$					
	(1)	(2)	(3)	(4)	(5)	(6)
Unc. $\times$ Infl. Surpr., AME	0.0183***	0.0092**	0.0145***	0.0059*	0.0181***	0.0099***
	[0.0102, 0.0264]	[0.0021, 0.0163]	[0.0073, 0.0217]	[−0.0004, 0.0122]	[0.0099, 0.0263]	[0.0025, 0.0173]
$\eta_{i,t}$ $\times$ Infl. Surpr., AME	−0.0205***	−0.0115***	−0.0164***	−0.0081**	−0.0205***	−0.0126***
	[−0.0286, −0.0124]	[−0.0187, −0.0043]	[−0.0236, −0.0092]	[−0.0145, −0.0017]	[−0.0287, −0.0123]	[−0.0200, −0.0052]
Other controls	Y	Y	Y	Y	Y	Y
CPI or PCE	CPI	CPI	PCE	PCE	PCE	PCE
Headline or core	C	C	H	H	C	C
Point or Mean	P	M	P	M	P	M
Observations	101,438	92,765	101,438	92,765	101,438	92,765

Note: Same as table 4, but varying the construction of the inflation surprise along three dimensions: (i) CPI vs. PCE inflation, (ii) headline vs. core inflation, (iii) mean of the belief distribution vs. respondents' point estimates. 95% confidence intervals based on standard errors that account for sampling uncertainty in the first stage and allow for clustering by both respondent and time are reported in brackets. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.6: Second Stage Regression: alternative dependent variable

	1 year inflation		3 year inflation	
	(1)	(2)	(3)	(4)
Unc. × Infl. Surpr., AME	0.0516**	0.0280***	0.0318*	0.0374***
	[0.0068, 0.0964]	[0.0178, 0.0382]	[−0.0003, 0.0639]	[0.0231, 0.0517]
$\eta_{i,t}$ × Infl. Surpr., AME	−0.0523**	−0.0308***	−0.0337**	−0.0400***
	[−0.0973, −0.0073]	[−0.0413, −0.0203]	[−0.0658, −0.0016]	[−0.0547, −0.0253]
Other controls	Y	Y	Y	Y
Type of infl. surpr.	Point	Mean	Point	Mean
Observations	80,822	99,984	78,356	97,263

Note: Analogous to table 4, but using the respondents' point estimate of inflation to construct the dependent variable. 95% confidence intervals based on standard errors that account for sampling uncertainty in the first stage and allow for clustering by both respondent and time are reported in brackets. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.7: Second Stage Regression: Additional Controls

	1 year inflation		3 year inflation	
	(1)	(2)	(3)	(4)
Unc. × Infl. Surpr., AME	0.0157***	0.0064**	0.0058*	−0.0030
	[0.0089, 0.0225]	[0.0000, 0.0128]	[−0.0005, 0.0121]	[−0.0084, 0.0024]
$\eta_{i,t}$ × Infl. Surpr., AME	−0.0314***	−0.0154	−0.0148*	−0.0008
	[−0.0468, −0.0160]	[−0.0357, 0.0049]	[−0.0303, 0.0007]	[−0.0182, 0.0166]
Other controls	Y	Y	Y	Y
Type of infl. surpr.	Point	Mean	Point	Mean
Observations	103,888	93,460	101,438	92,765

Note: Same as table 4, but with a larger set of controls: the Kronecker product of (inflation surprise; uncertainty×inflation surprise) with (a constant, the control function component ($\eta_{i,t}$), the inflation surprise, and $\eta_{i,t}^{inv}$), where $\eta_{i,t}^{inv}$ is the inverse of $(\eta_{i,t} - min(\eta_{i,t}) + 10)$. 95% confidence intervals based on standard errors that account for sampling uncertainty in the first stage and allow for clustering by both respondent and time are reported in brackets. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.8: Second Stage Regression: 2SLS Estimates (Robustness Across Inflation Measures)

(a) 1-Year Inflation

	Dep. var.: $\Delta m_{i,t}$							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Unc. × Infl. Surpr.	0.0143***	0.0116***	0.0214***	0.0160*	0.0199***	0.0155***	0.0282***	0.0206*
	[0.0086, 0.0200]	[0.0034, 0.0198]	[0.0097, 0.0331]	[−0.0011, 0.0331]	[0.0101, 0.0297]	[0.0038, 0.0272]	[0.0088, 0.0476]	[−0.0008, 0.0420]
Infl. Surpr.	−0.0866***	−0.0975**	−0.1874**	−0.1661	−0.1607**	−0.1409**	−0.3016**	−0.2212
	[−0.1488, −0.0244]	[−0.1898, −0.0052]	[−0.3485, −0.0263]	[−0.3831, 0.0509]	[−0.2903, −0.0311]	[−0.2797, −0.0021]	[−0.5942, −0.0090]	[−0.4956, 0.0532]
CPI or PCE	CPI	CPI	CPI	CPI	PCE	PCE	PCE	PCE
Headline or core	H	H	C	C	H	H	C	C
Point or Mean	P	M	P	M	P	M	P	M
Observations	103,888	93,460	103,888	93,460	103,888	93,460	103,888	93,460

(b) 3-Year Inflation

	Dep. var.: $\Delta m_{i,t}$							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Unc. × Infl. Surpr.	0.0094***	0.0032	0.0165**	0.0084**	0.0142**	0.0065	0.0216*	0.0105**
	[0.0028, 0.0160]	[−0.0023, 0.0087]	[0.0029, 0.0301]	[0.0003, 0.0165]	[0.0023, 0.0261]	[−0.0013, 0.0143]	[−0.0009, 0.0441]	[0.0010, 0.0200]
Infl. Surpr.	−0.0687*	−0.0166	−0.1766*	−0.0785	−0.1363	−0.0494	−0.2678	−0.1019*
	[−0.1461, 0.0087]	[−0.0769, 0.0437]	[−0.3722, 0.0190]	[−0.1747, 0.0177]	[−0.3002, 0.0276]	[−0.1382, 0.0394]	[−0.6225, 0.0869]	[−0.2180, 0.0142]
CPI or PCE	CPI	CPI	CPI	CPI	PCE	PCE	PCE	PCE
Headline or core	H	H	C	C	H	H	C	C
Point or Mean	P	M	P	M	P	M	P	M
Observations	101,438	92,765	101,438	92,765	101,438	92,765	101,438	92,765

Note: This table reports two-stage least squares (2SLS) estimates of the second-stage regression of $\Delta m_{i,t}$ (measured based on the bin mean) on the interaction of uncertainty and inflation surprises, Unc. $\times\, \widehat{\pi}_{i,t}$, and on the inflation surprise $\widehat{\pi}_{i,t}$. The endogenous regressor Unc. $\times\, \widehat{\pi}_{i,t}$ is instrumented using tenure interacted with the inflation surprise, i.e., $\text{Tenure}_{i,t} \times \widehat{\pi}_{i,t}$, with the inflation surprise included as an exogenous regressor. Panel (a) uses 1-year expectations, panel (b) uses 3-year expectations. Columns vary the construction of the inflation surprise along three dimensions: (i) CPI vs. PCE inflation, (ii) headline vs. core inflation, and (iii) point forecast vs. lagged bin mean. 95% confidence intervals based on standard errors two-way clustered by respondent and time are reported in brackets. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.